CUSTOMER BEHAVIOUR ANALYSIS

End-to-End Data Analytics Project (Python → MySQL → Power BI)

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1. Project Overview

This project analyzes customer purchasing behavior to extract actionable business insights across revenue, discounts, customer segmentation, product performance, and subscription patterns.

The workflow follows a complete analytics pipeline:

Data Cleaning (Python) → Data Storage (MySQL) → Analysis (SQL) → Visualization (Power BI)

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2. Tech Stack

- Python (Pandas, NumPy) – Data cleaning and preprocessing
- MySQL – Data storage and querying
- SQL – Analytical problem solving
- Power BI – Dashboard creation and visualization

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3. Data Preparation (Python)

- Cleaned raw dataset using Pandas
- Ensured correct data types:
    - Numerical: purchase_amount, review_rating, age
    - Categorical: gender, discount_applied, subscription_status, etc.
- Removed inconsistencies and validated dataset integrity
- Exported cleaned dataset for database ingestion

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4. Database Setup (MySQL)

- Created database:

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create database customer_behaviour;
use customer_behaviour;
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- Loaded cleaned dataset into table: customer_data
- Configured authentication and connection

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5. Business Questions and SQL Analysis

(All queries and findings derived from your SQL work)

Q1. Which category generates the highest revenue?

Finding:
Electronics category generates the highest revenue.

Business Impact:

- Helps prioritize high-performing categories
- Enables better inventory planning
- Improves marketing ROI

Q2. Are discounts increasing purchase value?

Findings:

- Average revenue is higher with discounts (₹219 vs ₹182)
- Total revenue remains lower due to low adoption
- Frequent buyers often purchase without discounts

Business Impact:

- Discounts should be targeted rather than broadly applied
- Helps avoid margin erosion
- Improves promotional efficiency

Q3. Revenue by gender

Finding:
Male customers generate the highest revenue.

Business Impact:

- Enables targeted marketing campaigns
- Improves customer segmentation
- Enhances personalization strategies

Q4. High-spending discount users

Finding:
Certain customers use discounts but still spend above average.

Business Impact:

- Identifies premium, price-sensitive customers
- Enables targeted discount campaigns
- Improves customer retention

Q5. Product rating analysis

Findings:

- Highest rated products: Gloves, Sandals
- Lowest rated products: Phone, Headphones

Business Impact:

- Promote high-rated products
- Improve low-performing products
- Enhance customer satisfaction

Q6. Shipping type vs average purchase

Findings:

- Express shipping has higher average purchase value
- Standard shipping has higher volume

Business Impact:

- Optimize shipping pricing strategies
- Upsell express delivery
- Increase average order value

Q7. Subscription vs revenue

Findings:

- Subscribers have higher average spend
- Non-subscribers contribute higher total revenue

Business Impact:

- Improve subscription adoption
- Strengthen loyalty programs
- Increase customer lifetime value

Q8. Discount-driven products

Top products with highest discount usage:
Laptop, Phone, Headphones, Watches

Business Impact:

- Identifies price-sensitive products
- Helps optimize discount strategies
- Reduces unnecessary margin loss

Q9. Customer segmentation

Segments created:

- New customers
- Returning customers
- Loyal customers

Business Impact:

- Enables personalized marketing
- Improves retention strategies
- Increases conversion rates

Q10. Repeat buyers vs subscription

Finding:
Repeat buyers exist but have relatively low subscription rates.

Business Impact:

- Highlights conversion gap
- Suggests need for better subscription value proposition
- Improves retention and engagement strategies

Q11. Top products within each category

Method:
Used CTE and window function (RANK) to identify top 3 products per category.

Business Impact:

- Improves product recommendations
- Supports inventory optimization
- Drives sales through best-selling products

Q12. Revenue contribution by age group

Finding:
Senior citizens contribute the highest revenue, followed by older adults.

Business Impact:

- Enables age-based targeting
- Improves marketing efficiency

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6. Power BI Dashboard

Developed an interactive dashboard to visualize insights.

Key Metrics

- Total Customers: 5000
- Total Revenue: ~994K
- Average Spend: ~198

- Average Review Rating: 3.63

Dashboard Sections

1. Business Overview
    - Total revenue
    - Total customers
    - Average spend
2. Customer Insights
    - Revenue by gender
    - Age group distribution
    - Subscription status
3. Product Insights
    - Top and bottom products by rating
    - Category-wise revenue
4. Behavioral Insights
    - Discount analysis
    - Purchase frequency
    - Shipping type vs spend
5. Geographic Insights
    - Top and bottom locations by revenue

Filters Implemented

- Category
- Gender
- Discount applied

- Subscription status

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7. Key Business Insights

- Discounts increase average order value but not total revenue
- High-value customers often do not depend on discounts
- Subscribers are more valuable but underrepresented
- Express shipping correlates with higher purchase value
- Senior citizens are a major revenue contributor
- Certain products are highly discount-sensitive
- Repeat buyers are not strongly converting to subscriptions

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8. Business Recommendations

- Implement targeted discount strategies
- Improve subscription offerings and incentives
- Focus on high-performing categories such as electronics
- Promote express shipping for high-value transactions
- Personalize marketing based on customer segments
- Focus on increasing customer lifetime value

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9. Project Strength

This project demonstrates:

- End-to-end data pipeline execution
- Advanced SQL usage (CTE, window functions, aggregations)
- Strong business-oriented analysis
- Effective data storytelling using Power BI